Sight Reduction Form

Vessel / Observer _____ Sight No. _____

Section I: Observations and Corrections		
1.	Celestial Body	
2.	**Apparent Altitude** Sextant Altitude (hs)	
	± Index Correction	
	− Dip Correction	
	Total Apparent Altitude (ha)	
3.	**Observed Altitude** ± Altitude Correction	
	± Additional Corrections (Atmospheric)	
	± Additional Corrections (Mars / Venus / Moon)	
	Total Observed Altitude (ho)	

Section II: Time and Dead Reckoning		
4.	Date (GMT)	
5.	DR Latitude	
6.	DR Longitude	
7.	Time (GMT)	

Section III: Local Hour Angle and Declination		
8.	**GHA** Tabulated GHA	
	+ GHA Increment	
	+ SHA (stars) *or* v-correction (Moon / planets, when applicable)	
	Total GHA	
9.	**LHA** (a) Choose Assumed Longitude so LHA minutes = 00.0′	
	W: a-minutes = GHA minutes, LHA = GHA−a. E: a-minutes = 60.0′−GHA minutes, LHA = GHA+a.	
	(b) Normalize: +360° if LHA < 0°; −360° if LHA ≥ 360°	
	Total LHA **(record normalized value, 0° to < 360°)**	
10.	**Declination** Tabulated Dec. (d-corr'n factor: _____)	
	± d-correction	
	Total Declination (Dec.)	
11.	**Assumed Latitude** Same / Contrary (compared to Dec. hemisphere)	

Section IV: Determining a Line of Position		
12.	**Computed Altitude** Tabulated hc (d: _____)	
	± Declination Increment (Dec. minutes / 60× d)	
	Total Computed Altitude (hc)	
13.	**Altitude Intercept** (ho or hc) whichever is *larger* ho = I.3	
	− (ho or hc) whichever is *smaller* hc = IV.12	
	(T / A) Intercept **(Toward if ho > hc, Away if ho < hc)**	
14.	**Azimuth Angle (Z)**	
15.	**Azimuth (Zn)**	

Zn from Z — quadrant rules

N Lat.: LHA > 180°: Zn = Z LHA < 180°: Zn = 360°−Z

S Lat.: LHA > 180°: Zn = 180°−Z LHA < 180°: Zn = 180°+Z

How to Fill Out the Sight Reduction Form

The form reduces a sextant reading to a *line of position* (LOP): a line somewhere on which the observer must lie. It compares the altitude you actually measured (**ho**) against the altitude you *would have* measured from an assumed position (**hc**). That difference, in nautical miles, is the **intercept**; the direction to the body is the **azimuth**. One sight gives one LOP; two or more crossing LOPs give a fix.

Angle notation. Enter angles as degrees and decimal minutes (e.g. 45° 13.4′). `HO229` takes the same thing compressed to `DD.MMm`: 45° 13.4′ is keyed `45.134`; LHA as `DDD.MMm`.

Section I — Observations and Corrections

1. Celestial body.
Sun (specify lower or upper limb), Moon, a planet, or a named star. This determines which correction tables apply later.

2. Apparent altitude.
Start with the raw sextant altitude **hs**. Apply the *index correction* for your sextant's own error (on the arc is subtractive, off the arc additive), then *dip* for your height of eye, which is always subtractive. The result is **ha**.

3. Observed altitude.
Apply the main altitude correction — refraction, semi-diameter, parallax — from the almanac's critical tables, entered with ha. Add the atmospheric correction if temperature or pressure are non-standard, and the extra correction for Mars, Venus, or the Moon where it applies. The result, **ho**, is the true altitude of the body's centre above the celestial horizon.

Section II — Time and Dead Reckoning

4–7. Date, DR position, time.
Record GMT (UT) to the second — four seconds of time is a nautical mile of longitude at the equator. The DR position is your best estimate at the moment of the sight; you round it to get the assumed position in items 9 and 11.

Section III — Local Hour Angle and Declination

8. GHA.
Take the tabulated GHA for the whole hour from the daily pages, add the increment for the minutes and seconds. For a star, add its SHA to the GHA of Aries; for the Moon, apply the v-correction.

9. LHA.
Choose the assumed longitude so the resulting LHA has 00.0′ of arc. West: give it the same minutes as GHA, then LHA = GHA − a. East: give it 60.0′−GHA minutes, then LHA = GHA + a. Normalize afterwards — add 360° if negative, subtract 360° if 360° or more. W and E are conditions telling you which operation to perform, not quantities to solve for.

10. Declination.
Tabulated declination for the hour, plus the d-correction for the minutes. Note carefully whether it is North or South.

11. Assumed latitude.
Round the DR latitude to a whole degree. Mark *Same* if the assumed latitude and the declination are in the same hemisphere, *Contrary* if opposite — the tables are laid out in those two cases and it is an easy place to go wrong.

Section IV — Determining a Line of Position

12. Computed altitude.
Enter Pub. 229 with LHA, assumed latitude, and declination; take out tabulated hc, d, and Z. Apply the declination increment (the odd minutes of declination, times d, divided by 60) to get **hc**.

13. Altitude intercept.
Subtract the smaller of ho and hc from the larger; one minute of arc is one nautical mile. Label it **T** (Toward) if ho is greater, **A** (Away) if hc is greater — *Ho Mo To*, ho more, toward.

14. Azimuth angle Z.
From the same table entry. Z is measured from the *elevated pole* — the pole nearer to you — and runs 0° to 180° east or west.

15. Azimuth Zn.
Convert Z to a true bearing 0°–360° using the quadrant rules boxed at the foot of page 1. This is the step the rules exist for: Z alone does not say which side of the meridian the body lies on, and in southern latitudes it is measured from the south pole rather than the north.

Plotting. From the assumed position, draw the azimuth line along Zn. Step off the intercept along it — toward the body or away as labelled — and mark that point. The LOP is the perpendicular to the azimuth through that point. Repeat with a second body; the crossing is your fix.

Using the `HO229` program in place of the tables

The program computes items 12 and 14 directly and does item 15 for you. Run it and enter, in `DD.MMm` notation:

`LAT N+ S-`
the assumed latitude from item 11; negative for South.

`LHA DDD.MMm`
the total LHA from item 9.

`DEC N+ S-`
the total declination from item 10; negative for South.

It returns `HC`, `Z`, and `ZN` in decimal degrees. `HC` goes straight into item 12 — there is no declination increment to apply, because the program uses the exact declination rather than a tabulated whole degree. `Z` is reported from the elevated pole, in the same sense as Pub. 229, so it goes in item 14 and the quadrant rules on page 1 apply to it unchanged. `ZN` is item 15, already converted — work item 15 by hand anyway the first few times and check that the two agree.

Because the program needs no whole-degree entering arguments, you may use your actual DR position and exact LHA instead of an assumed position — but then plot from the DR, consistently across every sight in a round. Sections I–III are unchanged: the program replaces the table lookup, and nothing else.